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//SIT 55 : ASSIGNMENT 6 : THREADED BINARY TREE
/*Implement In-order Threaded Binary Tree and traverse it in
In-order and Pre-order.*/

#include<iostream>
using namespace std;
class node
{
 public:
 int data;
 int lth,rth;
 node *left,*right;
};
class thread
{
 private:
 node *dummy;
 node *New,*root,*temp,*parent;
 public:
 thread();
 void create();
 void insert(node *,node *);
 void display();
 void find();
 void delet();
 void del(node *root,node *dummy,int key);
};
thread::thread()
{
 root=NULL;

}
void thread::create()
{
 New=new node;
 New->left=NULL;
 New->right=NULL;
 New->lth=0;
 New->rth=0;
 cout<<"\nEnter The Element: ";
 cin>>New->data;
 if(root==NULL)
 {
 root=New;
 dummy=new node;
 dummy->data=-999;
 dummy->left=root;
 root->left=dummy;
 root->right=dummy;
 }
 else
 insert(root,New);
}
void thread::insert(node *root,node *New)
{
 if(New->data<root->data)
 {
 if(root->lth==0)
 {
 New->left=root->left;
```

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 New->right=root;
 root->left=New;
 root->lth=1;
 }
 else
 insert(root->left,New);

 }
 if(New->data>root->data)
 {
 if(root->rth==0)
 {
 New->right=root->right;
 New->left=root;
 root->rth=1;
 root->right=New;
 }
 else
 insert(root->right,New);

 }
}
void thread::display()
{
 void inorder(node*,node *dummy);
 void preorder(node*,node *dummy);
 if(root==NULL)
 cout<<"Tree is not created.";
 else
 {
 cout<<"\nInorder Traversal: ";
 inorder(root,dummy);
 cout<<"\nPreorder Traversal: ";
 preorder(root,dummy);

 }

}
void inorder(node *temp,node *dummy)
{
 while(temp!=dummy)
 {
 while(temp->lth==1)
 temp=temp->left;
 cout<<" "<<temp->data;
 while(temp->rth==0)
 {
 temp=temp->right;
 if(temp==dummy)
 return;
 cout<<" "<<temp->data;

 }
 temp=temp->right;

 }
}
void preorder(node *temp,node *dummy)
{
 int flag=0;
```

```
 while(temp!=dummy)
 {
 if(flag==0)
 cout<<" "<<temp->data;
 if((temp->lth==1)&&(flag==0))
 {
 temp=temp->left;

 }
 else
 {
 while(1)
 {
 if(temp->rth==1)
 {
 flag=0;
 temp=temp->right;
 break;

 }
 else
 {
 if(temp!=dummy)
 {
 flag=1;
 temp=temp->right;
 break;

 }
 }
 }
 }
 }
}
int main()
{
 int choice;
 char ans='N';
 thread th;
 do
 {
 cout<<"Program For Threaded Binary Tree\n";
 cout<<"\n1)Create \n2)Display";
 cout<<"\nEnter Your Choice: ";
cin>>choice;
 switch(choice)
 { case 1:
 do
 {
 th.create();
 cout<<"\nDo you want to enter more elements?(y/n): ";
 cin>>ans;
 }while(ans=='y');
 break;
 case 2:th.display();
 break;}cout<<"\n\nWant To See Main Menu?(y/n):
";cin>>ans;}while(ans=='y');
 return 0;
}
```

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/*OUTPUT:
Program For Threaded Binary Tree

1)Create
2)Display
Enter Your Choice: 1
Enter The Element: 66
Do you want to enter more elements?(y/n): y
Enter The Element: 92
Do you want to enter more elements?(y/n): y
Enter The Element: 21
Do you want to enter more elements?(y/n): y
Enter The Element: 4
Do you want to enter more elements?(y/n): n
Want To See Main Menu?(y/n): y
Program For Threaded Binary Tree

1)Create
2)Display
Enter Your Choice: 2
Inorder Traversal:  4 21 66 92
Preorder Traversal:  66 21 4 92*/
```